

Passing or Rushing: Who Wins Games in the NFL?

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Motivation

- Modern NFL is often called a “passing league”
- But rushing performance may still affect winning
- Goal:
 - Compare passing vs rushing efficiency
 - Analyze which offensive style is more related to winning

Main Dataset:

- nfl_data_py API
- Official NFL play-by-play data
- Seasons: 2022 – 2024
- Only regular season games included

Additional References:

- Pro-Football-Reference
- Kaggle NFL datasets

Feature Engineering

Custom offensive metrics were created:

- **PEI (Passing Efficiency Index)**

$$\frac{\text{Passing Yards}}{\text{Pass Attempts}}$$

- **REI (Rushing Efficiency Index)**

$$\frac{\text{Rushing Yards}}{\text{Rush Attempts}}$$

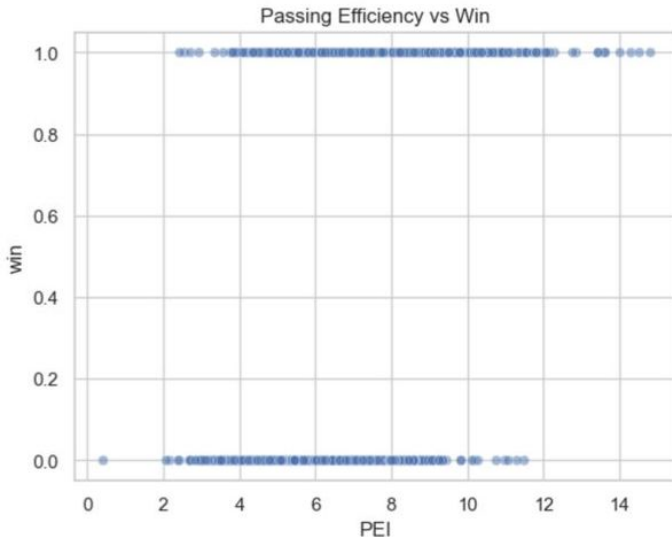
- **OBR (Offensive Balance Ratio)**

$$\frac{\text{Pass Attempts}}{\text{Pass Attempts} + \text{Rush Attempts}}$$

Data Processing Pipeline

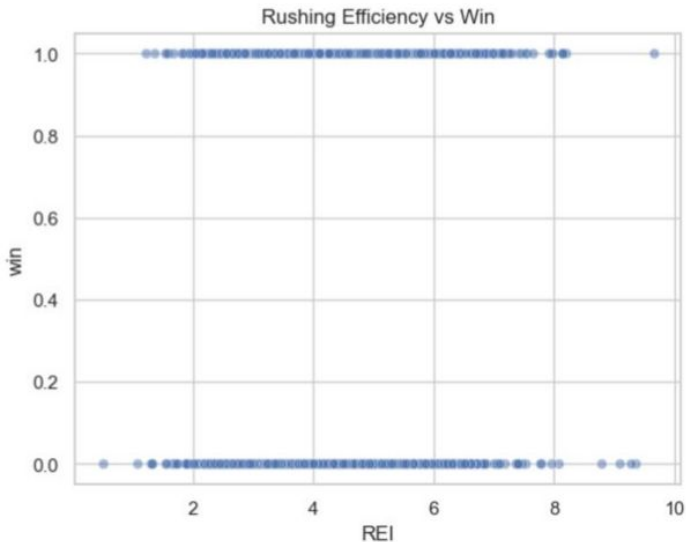
- 1 Import play-by-play NFL data
- 2 Filter regular season games
- 3 Aggregate team-level game statistics
- 4 Create PEI, REI, and OBR metrics
- 5 Merge with game outcomes
- 6 Perform EDA and statistical analysis
- 7 Train Logistic Regression model

Passing Efficiency vs Winning



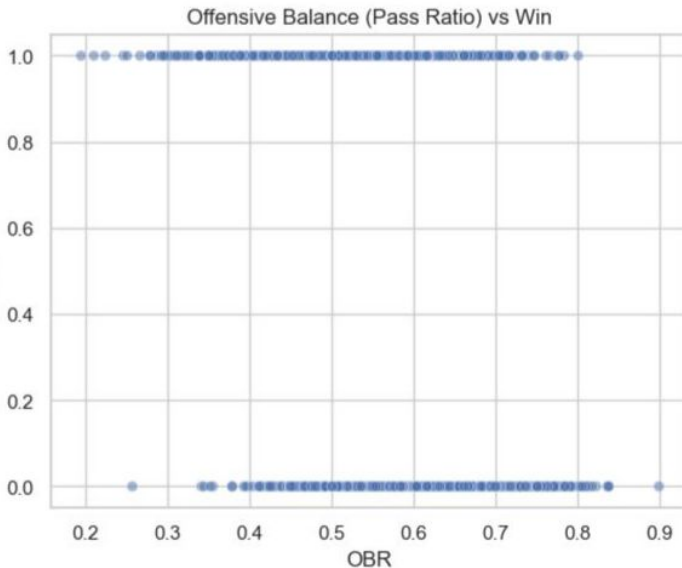
Higher PEI values are more commonly associated with winning teams.

Rushing Efficiency vs Winning

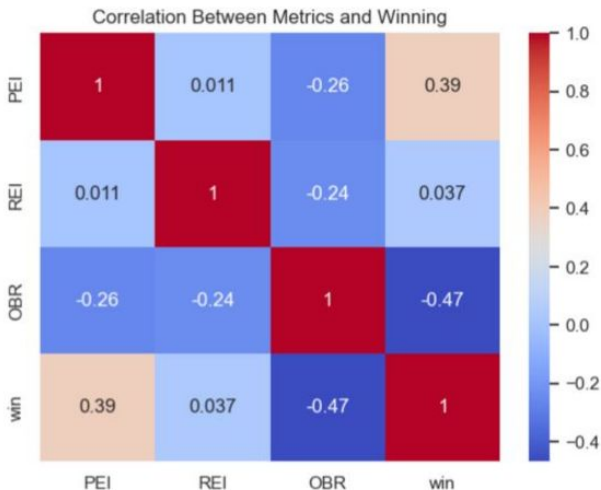


REI does not show a strong separation between wins and losses.

Offensive Balance Ratio vs Winning



Correlation Heatmap



PEI has the strongest positive correlation with winning, while OBR shows a negative relationship with wins.

Hypothesis Testing

Metric	t-statistic	p-value
PEI	17.20	7.0×10^{-61}
REI	1.49	0.136
OBR	-21.28	7.5×10^{-89}

Interpretation:

- PEI and OBR are statistically significant
- REI is not statistically significant
- OBR may reflect game situation effects rather than offensive superiority

Machine Learning Model

Model Used:

- Logistic Regression

Input Features:

- PEI
- REI
- OBR

Results:

- Accuracy: **76.7%**
- PEI had the strongest positive coefficient
- REI had weak influence

Important Limitation

- This project mainly measures correlations, not causation
- Winning teams may pass efficiently because they are already strong teams
- Some variables may also be affected by game situation
- Example:
 - Teams leading late in games often run the ball more
 - This can reduce OBR even for winning teams

Main Findings

- Passing efficiency is strongly associated with winning
- Rushing efficiency alone is not a strong predictor
- OBR results may partly reflect late-game strategy effects
- Logistic Regression predicts outcomes with around 77% accuracy

Conclusion: Passing-related metrics appear to be more informative than rushing efficiency for predicting NFL game outcomes.

Limitations

- Defensive metrics not included
- Weather and injuries ignored
- Simplified offensive metrics
- Correlation does not imply causation

Future Work

- Add defensive statistics
- Use more NFL seasons
- Try advanced ML models
- Include score difference and game context

Thank You